THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number is divisible by both 5 and 7. It must be divisible by: 1

(a) 40 **(b)** 12 **(c)** 70 **(d)** 35

2. The smallest number which leaves remainder 9 when divided by 16 and 24 is: 1

(a) 57 **(b)** 393 **(c)** 48 **(d)** 105

3. The number of multiples of 25 between 300 and 2000 (both included) is: 1

(a) 71 **(b)** 68 **(c)** 69 **(d)** 70

4. The number of 3-digit numbers divisible by 22 is: 1

(a) 42 **(b)** 41 **(c)** 45 **(d)** 40

5. The smallest 3-digit palindrome divisible by 7 is: 1

(a) 151 **(b)** 168 **(c)** 161 **(d)** 171

6. If the number 4a6b is divisible by 90, then $a + b$ equals: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

7. The number of steps the Kaprekar routine needs to reach 6174 starting from 1004 is: 1

(a) 9 **(b)** 6 **(c)** 7 **(d)** 8

8. A number is divisible by both 7 and 8. It must be divisible by: 1

(a) 56 **(b)** 15 **(c)** 63 **(d)** 112

9. **Assertion (A):** The number 2,728 is divisible by 11. 1

Reason (R): A number is divisible by 11 if the difference between the sum of the digits at odd places and the sum of the digits at even places is 0 or a multiple of 11.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Applying Kaprekar's routine to 3524 (largest arrangement minus smallest arrangement, repeatedly) leads to 6174. 1
Reason (R): Kaprekar's routine on any four-digit number whose digits are not all equal always ends at 495.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sum of five consecutive numbers is 115. Find the numbers. 2

12. Find the smallest number which leaves remainder 8 when divided by 30 and 45. 2

13. How many multiples of 125 lie between 1000 and 5000 (both included)? 2

14. The sum of five consecutive numbers is 150. Find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the six-digit number 683683 is divisible by 7, 11 and 13. 3

16. Find the greatest 4-digit number divisible by 16, 20 and 30. 3

17. How many 3-digit numbers are divisible by both 12 and 21? 3

OR

Show that the six-digit number 791791 is divisible by 7, 11 and 13. 3

18. Find the greatest 4-digit number divisible by 12, 18 and 27. 3

OR

How many 3-digit numbers are divisible by both 16 and 18? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Aman learns about palindromic numbers, which read the same from left to right and right to left. He also tries the reverse-and-add routine on 57. 5
(i) Is 12,321 a palindrome?
(ii) Find the sum of 57 and its reverse.
(iii) Repeat the process on that sum; what is obtained?
(iv) Show that every four-digit palindrome $abba$ is divisible by 11, and verify with 3,663.

20. Find the digits a and b if the number 2a5b is divisible by 18. List all possibilities. 5

OR

Take the number 971. Subtract its reverse. Add the reverse of the result to the result. What do you get? Show that this always happens for 3-digit numbers whose first and last digits differ by 2 or more. 5

END OF QUESTION PAPER — **Total: 40 Marks**